

Continuous Bivariate Distributions



Carl Gustav Jacob Jacobi

Carl Gustav Jacob Jacobi (10 December 1804 – 18 February 1851) was a German mathematician, who made fundamental contributions to elliptic functions, dynamics, differential equations, and number theory. His name is occasionally written as **Carolus Gustavus Iacobus Iacobi** in his Latin books, and his first name is sometimes given as Karl.

Jacobi was the first Jewish mathematician to be appointed professor at a German university.

Contents ...

- 2.0 Introduction
- 2.1 Continuous Bivariate R.V.
- 2.2 Joint Probability Density Function
- 2.3 Distribution Function
- 2.4 Marginal Probability Distributions
- 2.5 Conditional Probability Distributions
- 2.6 Independence of Random Variables
- 2.7 Expectation, Mean and Moments
- 2.8 M.G.F.
- 2.9 Conditional Expectation
- 2.10 Transformations of Bivariate Continuous Random Variables

Key Words :

Continuous bivariate r.v., Joint p.d.f., Marginal and conditional probability distributions, Conditional expectations, Joint M.G.F, Transformation of (X, Y).

Objectives :

To find the marginal probability distribution and conditional probability distributions. Identify whether the variables are independent. To find the joint moments and M.G.F. To find probabilities of joint events of X and Y . To find correlation and regression lines. To find the probability distributions of functions of (X, Y) .

2.0 Introduction

Suppose Ω is a uncountably infinite sample space. We can associate two real valued function with Ω . For example, in a manufacturing process inspection of spring is conducted. We record inner diameter as well as tensile strength of each spring. Thus, with respect to every sample point we record values of two jointly variable quantities. The joint study of such variables needs bivariate probability distribution to be introduced. Moreover, we require to find the distribution of sum of two or more random variables. In this situation, we need to use bivariate or multivariate probability distribution. In this chapter we particularly study bivariate continuous type of probability distributions.

2.1 Continuous Bivariate Random Variable

Definition : Let, Ω be the sample space corresponding to a random experiment. Let, $X(\omega)$ and $Y(\omega)$ be two real valued continuous functions with domain Ω . Then the ordered pair $(X(\omega), Y(\omega))$ is called a bivariate two-dimensional continuous random variable, ω being sample point in Ω .

Note :

- (1) For simplicity we write X and Y in place of $X(\omega)$ and $Y(\omega)$ respectively.
- (2) If X has range set $A \subset \mathbb{R}$ and Y has range set $B \subset \mathbb{R}$ then,
 $(X, Y) : \Omega \rightarrow A \times B$.

Illustrations :

- (1) $X(\omega)$ = Weight of ω^{th} student.
 $Y(\omega)$ = Height of ω^{th} student.
- (2) $X(\omega)$ = Distance of a space denoted by ω from equator.
 $Y(\omega)$ = Average temperature at the same place ω .
- (3) $X(\omega)$ = pH value (acidity) of water at a pond at time ω .
 $Y(\omega)$ = Growth of fungal plant at the same time ω .

2.2 Joint Probability Density Function

Earlier in 1.2 we have seen that in case of continuous random variable, probability mass is not attached to any individual value. If (X, Y) is a bivariate random variable then probability is attached to a two dimensional region, such as $\{(x, y) \mid a < x < b, c < y < d\}$. The probability varies from region to region. In order to obtain the probability mass associated with any two dimensional region we need to take into account the concept of probability density function. A bivariate function $f(x, y)$ can be treated as probability density function if, it is non-negative and continuous function of (x, y) . The probability that a r.v. (X, Y) takes values in a small region

$$\left\{ (x, y) \mid x - \frac{\delta x}{2} < X < x + \frac{\delta x}{2}, y - \frac{\delta y}{2} < Y < y + \frac{\delta y}{2} \right\}$$

will be the product of area of region and value of $f(x, y)$ in the region.

$$\therefore P\left(x - \frac{\delta x}{2} < X < x + \frac{\delta x}{2}, y - \frac{\delta y}{2} < Y < y + \frac{\delta y}{2}\right) \simeq f(x, y) \delta x, \delta y.$$

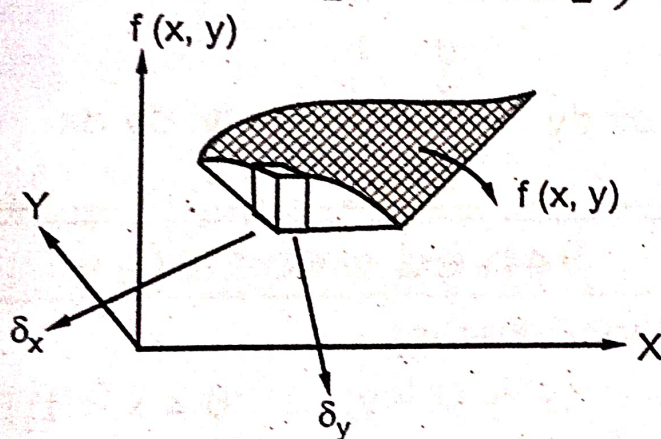


Fig. 2.1

Thus, the above probability is a volume of a box with dimensions $f(x, y)$, δx and δy .

Definition : A function $f(x, y)$ is called a joint probability density function of a bivariate continuous r.v. (x, y) if,

$$(i) \quad f(x, y) \geq 0; -\infty < x, y < \infty$$

$$(ii) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1.$$

$$\text{Note: (1) } P(a < x < b, c < y < d) = \int_a^b \int_c^d f(x, y) dx dy$$

= A volume of box with rectangular base having sides $(b - a)$ and $(d - c)$ and top surface given by $f(x, y)$.

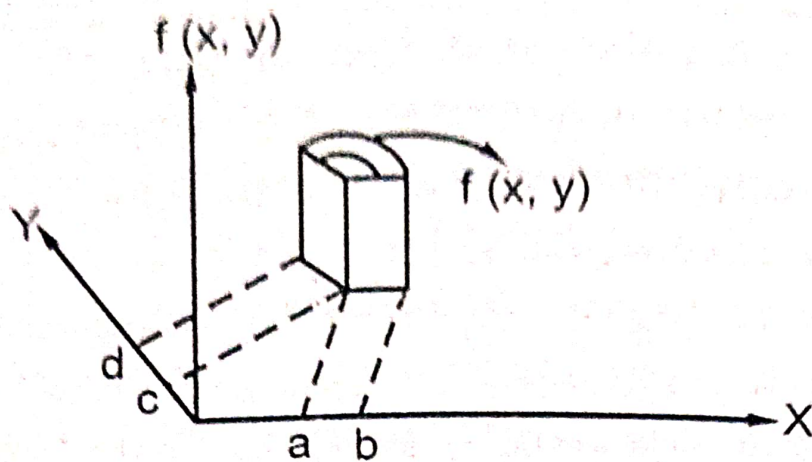


Fig. 2.2

$$(2) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$$

The total volume under surface $f(x, y)$ which is bounded by xy -plane = 1.

(3) If a, b, c, d are constants then,

$$\int_a^b \int_c^d f(x, y) dx dy = \int_c^d \int_a^b f(x, y) dy dx$$

Solved Examples

Example 2.1 : Verify whether,

$$f(x, y) = 12xy(1-y) \quad ; \quad 0 < x < 1; 0 < y < 1$$

$$= 0 \quad ; \quad \text{otherwise}$$

is a joint p.d.f. of (X, Y) .

Solution : Since, $x \geq 0, y \geq 0, 1 - y \geq 0$,

$$f(x, y) = 12xy(1-y) \geq 0$$

Let, us evaluate $\int_0^1 \int_0^1 f(x, y) dy dx$

$$= \int_0^1 \int_0^1 12xy(1-y) dy dx = 12 \int_0^1 x \left[\int_0^1 y(1-y) dy \right] dx$$

$$= 12 \int_0^1 x \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 dx = 12 \frac{1}{6} \int_0^1 x dx = 2 \cdot \left[\frac{x^2}{2} \right]_0^1 = 1$$

$\therefore f(x, y)$ is a joint p.d.f. of (X, Y) .

Example 2.2 : If $f(x, y)$ is a p.d.f. Find the value of constant c ; where,
 $f(x, y) = c$; for $x > 0, y > 0, x + y < 2$.
 $= 0$; otherwise.

Solution : (i) $f(x, y) \geq 0$ implies $c \geq 0$

$$(ii) \quad \iint f(x, y) dy dx = 1$$

$$\therefore \int_{x=0}^2 \int_{y=0}^{2-x} c dy dx = 1$$

$$\therefore c \int_0^2 [y]_0^{2-x} dx = 1$$

$$c \int_0^2 (2-x) dx = 1$$

$$c \left[2x - \frac{x^2}{2} \right]_0^2 = 1$$

$$c [4 - 2] = 2c = 1$$

$$\therefore c = \frac{1}{2}$$

$$\therefore f(x, y) = \frac{1}{2} \quad ; \quad \text{for } x > 0, y > 0, x + y < 2$$

$$= 0 \quad ; \quad \text{otherwise.}$$

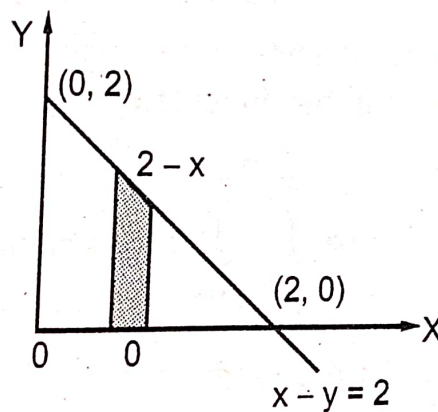


Fig. 2.3

Double Integrals : While dealing with continuous random variables we need to evaluate double integrals many times. We summarize the required results in the subsequent discussion.

Note : (1) To evaluate $\iint f(x, y) dx dy$, first we evaluate inner integral. Here, dx appears first, so inner integral is w.r.t. x and outer is w.r.t. y . Exactly opposite will be about $\iint f(x, y) dy dx$. Here, inner integral is w.r.t. y where as outer integral is w.r.t. x .

(2) If $\int_c^d \int_a^b f(x, y) dx dy$, then inner integral is w.r.t. x , over the range a to b and outer integral is w.r.t. y , over the range (c, d) .

(3) If a, b, c, d are constants then, order of integration is immaterial.

$$\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

i.e. order of integration can be interchanged.

(4) If limits of integration w.r.t. y are functions of x and those of x are constant then inner integral is w.r.t. y .

$$\int_c^d \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

On the other hand if the limits of integration w.r.t. X are functions of Y and those of Y are constants then inner integral is w.r.t. X and other w.r.t. Y .

$$\int_a^b \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

Thus, the limits of outer integral are always constant.

(5) Suppose A is a region of integration given by,

$A = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$. If we can express A such that the limits of x as functions of y and limits of y constant i.e. $A = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$ then, order of integral can be interchanged as follows:

$$\int_A f(x, y) dy dx = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

Example 2.3 : If (x, y) is a r.v. with joint p.d.f.

$$f(x, y) = e^{-(x+y)}; x \geq 0, y \geq 0.$$

Find $P(X + Y \leq 1)$.

Solution : Let, $A = \{(x, y) \mid x + y \leq 1, x \geq 0, y \geq 0\}$

$$\text{To evaluate, } P(X + Y \leq 1) = \iint_A f(x, y) dx dy.$$

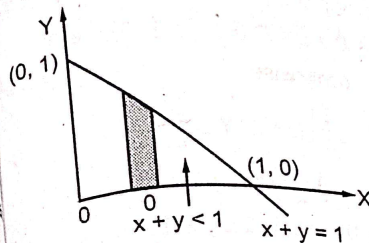


Fig. 2.4

Thus, $A = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$.

Alternative Approach :

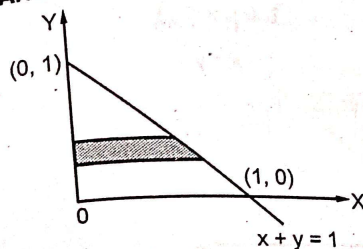


Fig. 2.5

Thus, the same region A will be $\{(x, y) \mid 0 \leq y \leq 1, 0 \leq x \leq 1 - y\}$

$$\begin{aligned} \therefore P(A) &= \int_0^1 \int_0^{1-y} e^{-(x+y)} dy dx \text{ or } P(A) \\ &= \int_0^1 \int_0^{1-y} e^{-(x+y)} dx dy = \int_0^1 e^{-y} \left[\frac{e^{-x}}{-1} \right]_0^{1-y} dy \\ &= \int_0^1 e^{-y} [-e^{-(1-y)} + 1] dy \\ &= \int_0^1 (e^{-y} - e^{-1}) dy = \int_0^1 e^{-y} dy - e^{-1} \int_0^1 dy \\ &= 1 - e^{-1} - e^{-1} = 1 - 2e^{-1} = 0.2642 \end{aligned}$$

Example 2.4 : In a r.v. (X, Y) has joint p.d.f.
 $f(x, y) = x + y$; $0 \leq x \leq 1, 0 \leq y \leq 1$
 $= 0$; otherwise.

Then, find (i) $P(X \leq Y)$, (ii) $P(X^2 + Y^2 \leq \frac{1}{4})$ (iii) $P(Y \geq \frac{1}{2})$.

Solution : (i) To evaluate $P(X \leq Y)$, we denote region
 $A = \{(x, y) | x \leq y, 0 \leq x \leq 1, 0 \leq y \leq 1\}$.

We need to specify the limit of region a separately in terms of variable.

$$A = \{(x, y) | 0 \leq x \leq 1, x \leq y \leq 1\}$$

$$\text{Alternatively, } A = \{(x, y) | 0 \leq y \leq 1, 0 \leq x \leq y\}$$

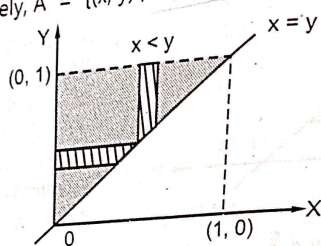


Fig. 2.6

$$P(X \leq Y) = \iint_A f(x, y) dx dy \quad \left(\text{or} \iint_A f(x, y) dy dx \right)$$

$$\begin{aligned} \therefore \int_0^1 \int_0^y x + y dx dy &= \int_0^1 \left[\frac{x^2}{2} + xy \right]_0^y dy \\ &= \int_0^1 \left[\frac{y^2}{2} + y^2 \right] dy = \frac{3}{2} \left[\frac{y^3}{3} \right]_0^1 = \frac{1}{2} \end{aligned}$$

(ii) To evaluate $P(X^2 + Y^2 \leq \frac{1}{4})$ we denote

$$B = \{(x, y) | x^2 + y^2 \leq \frac{1}{4}, 0 \leq x \leq \frac{1}{2}, 0 \leq y \leq \frac{1}{2}\}$$

The region B in terms of x and y, separately can be written as follows.

$$B = \{(x, y) | 0 \leq x \leq \frac{1}{2}, 0 \leq y \leq \sqrt{\frac{1}{4} - x^2}\}$$

$$B = \{(x, y) | 0 \leq y \leq \frac{1}{2}, 0 \leq x \leq \sqrt{\frac{1}{4} - y^2}\}$$

or

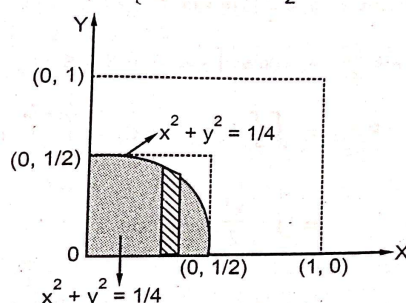


Fig. 2.7

$$\begin{aligned} P(B) &= \iint_B f(x, y) dy dx = \int_0^{1/2} \int_0^{\sqrt{\frac{1}{4} - x^2}} (x + y) dy dx \\ &= \int_0^{1/2} \left[xy + \frac{y^2}{2} \right]_0^{\sqrt{\frac{1}{4} - x^2}} dx \\ &= \int_0^{1/2} x \sqrt{\frac{1}{4} - x^2} dx + \frac{1}{2} \int_0^{1/2} (\frac{1}{4} - x^2) dx \\ &= I_1 + I_2 \end{aligned}$$

To evaluate I_1 put $t = \frac{1}{4} - x^2$

$\therefore dt = -2x dx$ and for $x = 0, t = \frac{1}{4}$, for $x = \frac{1}{2}, t = 0$

$$\therefore I_1 = \int_0^{1/2} x \sqrt{\frac{1}{4} - x^2} dx = \frac{1}{2} \int_{1/4}^0 t^{1/2} dt = \frac{1}{2} \left[\frac{t^{3/2}}{3/2} \right]_{1/4}^0 = \frac{1}{24}$$

and $I_2 = \int_0^{\frac{1}{2}} \frac{1}{4} - x^2 dx = \left[\frac{1}{4}x - \frac{x^3}{3} \right]_0^{\frac{1}{2}} = \frac{1}{8} - \frac{1}{24} = \frac{2}{24}$

$\therefore P(B) = I_1 + I_2 = \frac{1}{24} + \frac{1}{24} = \frac{1}{12}$

(iii) To evaluate $P\left(y \geq \frac{1}{2}\right)$ we denote

$$C = \left\{ y \mid y \geq \frac{1}{2} \right\} = \left\{ (x, y) \mid y \geq \frac{1}{2}; 0 \leq x \leq 1 \right\}$$

$$\begin{aligned} P(C) &= \int_C \int f(x, y) dx dy = \int_{1/2}^1 \int_0^1 (x + y) dx dy \\ &= \int_{1/2}^1 \left[\frac{x^2}{2} + xy \right]_0^1 dy \\ &= \int_{1/2}^1 \left[\frac{1}{2} + y \right] dy = \left[\frac{y}{2} + \frac{y^2}{2} \right]_{1/2}^1 = 1 - \frac{3}{8} = \frac{5}{8} \end{aligned}$$

2.3 Distribution Function

Definition : If (X, Y) is a bivariate r.v. with joint p.d.f. $f(x, y)$, joint distribution function is denoted by $F(x, y)$ and defined as,

$$F(x, y) = P(X \leq x; Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) dv du \text{ for } x \in R \text{ and } y \in R$$

Remark : $F(x, y) : R \times R \rightarrow [0, 1]$

Example 2.5 : If (X, Y) is a bivariate r.v. with p.d.f.
 $f(x, y) = \frac{e^{-(x+y)}}{2}; x \geq 0, y \geq 0$
 $= 0; \text{ otherwise.}$

Find distribution function of (X, Y) .

Solution : By definition,

$$F(a, b) = P(X \leq a, Y \leq b)$$

$$\begin{aligned} &= \int_0^a \int_0^b e^{-(x+y)} dy dx = \int_0^a e^{-x} \left[\int_0^b e^{-y} dy \right] dx \\ &= \int_0^a e^{-x} (1 - e^{-b}) dx = (1 - e^{-b}) (1 - e^{-a}). \end{aligned}$$

Properties of Distribution Function $F(x, y)$

(1) **Non-negative :** $F(x, y) \geq 0, \forall (x, y) \in R \times R$.

Proof : $F(x, y) = P(X \leq x, Y \leq y)$

Since, $F(x, y)$ is a probability, it is non-negative. Moreover, it lies between $[0, 1]$.

(2) **Non-decreasing in each argument :**

Statement : If $x_1 < x_2$, then $F(x_1, y) < F(x_2, y)$ and if $y_1 < y_2$, then $F(x, y_1) < F(x, y_2)$

Proof : Since $x_1 < x_2$ we get

$$\{(x, y) \mid x \leq x_1, Y \leq y\} \subset \{(x, y) \mid x \leq x_2, Y \leq y\}$$

$$\therefore P(X \leq x_1, Y \leq y) \leq P(X \leq x_2, Y \leq y)$$

$$\therefore F(x_1, y) \leq F(x_2, y)$$

Similarly, it can be proved that $F(x, y_1) \leq F(x, y_2)$ for $y_1 < y_2$.

(3) (a) $F(-\infty, -\infty) = 0$

(b) $F(\infty, \infty) = 1$

(c) $F(x, -\infty) = F(-\infty, y) = 0$

Proof : (a) $F(-\infty, -\infty) = \lim_{\substack{x \rightarrow -\infty \\ y \rightarrow -\infty}} F(x, y) = P(X \leq -\infty, Y \leq -\infty) = 0$

(b) $F(\infty, \infty) = \lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} F(x, y) = P(X \leq \infty, Y \leq \infty) = 1$

(c) $F(x, -\infty) = \lim_{y \rightarrow -\infty} F(x, y) = P(X \leq x, Y \leq -\infty) = 0$

Similarly, $F(-\infty, y) = 0$

(4) **Right-Continuity :** $F(x, y)$ is right continuous in each of its argument :

$$\therefore \lim_{h \rightarrow 0} F(x + h, y) = F(x, y); \text{ for } y \text{ fixed and } h > 0$$

$$\lim_{k \rightarrow 0} F(x, y + k) = F(x, y); \text{ for } x \text{ fixed and } k > 0$$

(5) If $a < b$ and $c < d$ then,

$$\Delta^2 F(x, y) = F(b, d) - F(a, d) - F(b, c) + F(a, c) \geq 0.$$

Proof : Note that,

$$\begin{aligned} P(a < X \leq b, c < Y \leq d) &\geq 0 \text{ being a probability} \\ &= P(X \leq b, Y \leq d) - P(X \leq a, Y \leq d) - P(X \leq b, Y \leq c) + P(X \leq a, Y \leq c) \\ &= F(b, d) - F(a, d) - F(b, c) + F(a, c) \geq 0. \end{aligned}$$

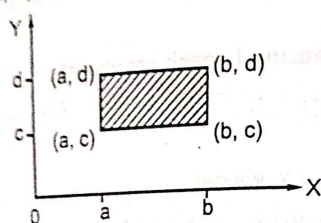


Fig. 2.8

Remark :

(1) $F(x, \infty) = \lim_{y \rightarrow \infty} F(x, y) = F_1(x)$ = distribution function of X

$F(\infty, y) = \lim_{x \rightarrow \infty} F(x, y) = F_2(y)$ = distribution function of Y .

(2) If $F(x, y)$ is a distribution function twice differentiable, then,

joint p.d.f. is $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$

2.4 Marginal Probability Distributions

If (X, Y) is a continuous bivariate r.v. with joint p.d.f. $f(x, y)$, then the probability distribution of X alone is called as marginal probability distribution of X and the probability distribution of Y alone is called as marginal probability distribution of Y .

The marginal p.d.f. of x is given by,

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

and the marginal p.d.f. of Y is given by

$$f_2(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

Thus, to get marginal p.d.f. of a variables we integrate joint p.d.f. over range of the other variable.

Note : Some times among the two variables (X, Y) range of X is constant and that of Y is a function of X or the range of Y is constant and that of X is a function of Y .

While obtaining marginal p.d.f. of X we need to take range of X constant (free from Y). Similarly to obtain the marginal p.d.f. of Y we have to express the range of Y constant (i.e. free from X). Illustration 6 will demonstrate the procedure.

2.5 Conditional Probability Distributions

If a continuous r.v. (X, Y) has joint p.d.f. $f(x, y)$, then the conditional p.d.f. of X given $Y = y$ is,

$$g_1(x | Y = y) = \frac{f(x, y)}{f_2(y)} ; \text{ if } f_2(y) > 0$$

Similarly, the conditional p.d.f. of Y given $X = x$ is

$$g_2(y | X = x) = \frac{f(x, y)}{f_1(x)} ; \text{ if } f_1(x) > 0$$

where $f_1(x)$ and $f_2(y)$ are marginal p.d.f.s of r.v. X and Y respectively.

Note : Sometimes the range of X is constant and that of Y is function of X , or range of Y is constant and that of X is function of Y . To obtain the conditional p.d.f. of X we need to find the range of X expressed in terms of Y (i.e. function of Y). Similarly, to obtain the conditional distribution of Y , we find the range of Y expressed in terms of X (i.e. function of X). This process is exactly opposite to that of used in getting marginal p.d.f.

Example 2.6 : Let $f(x, y) = 21x^2y^3$, $0 < x < y < 1$
 $= 0$, otherwise

be the joint p.d.f. of r.v. (X, Y) .

- Find marginal distributions of X, Y .
- Find conditional distribution of X given $Y = y$ and that of Y given $X = x$.

Solution : The range set

$A = \{(x, y) | 0 < x < y < 1\}$ can be written so that limits of X will be constants and the limits of Y will be functions of X as follows :

$$A_1 = \{(x, y) | 0 < x < 1, x < y < 1\}$$

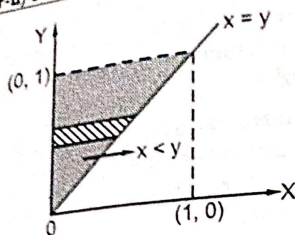


Fig. 2.9

Similarly, we can choose limits of Y constants and limits of X functions of Y .

$A_2 = \{(x, y) | 0 < y < 1, 0 < x < y\}$. Note that $A = A_1 = A_2$.

(i) To find marginal distribution of X we have to use A_1

$$\begin{aligned} f_1(x) &= \int_x^1 f(x, y) dy \quad ; \quad 0 \leq x \leq 1 \\ &= \int_x^1 21x^2y^3 dy = 21x^2 \left[\frac{y^4}{4} \right]_x^1 \\ &= \frac{21}{4} x^2 (1 - x^4) \quad ; \quad 0 \leq x \leq 1 \end{aligned}$$

To find the marginal distribution of Y , we have to use range set A_2

$$f_2(y) = \int_0^y f(x, y) dx = \int_0^y 21x^2y^3 dx = 21y^3 \left[\frac{x^3}{3} \right]_0^y = 7y^6, \quad 0 \leq y \leq 1.$$

(ii) To find the conditional p.d.f. of X given $Y = y$ we have to use A_1 in which limits of X are functions of Y .

$$\begin{aligned} g_1(x|y) &= \frac{f(x, y)}{f_2(y)} ; \quad 0 \leq x \leq y \\ &= \frac{21x^2y^3}{7y^6} = \frac{3x^2}{y^3} \quad 0 \leq x \leq y. \end{aligned}$$

Similarly, conditional p.d.f. of Y given $X = x$ is,

$$\begin{aligned} g_2(y|x) &= \frac{f(x, y)}{f_1(x)} ; \quad x \leq y \leq 1 \\ &= \frac{21x^2y^3}{\frac{21}{4}x^2(1-x^4)} = 4 \frac{y^3}{(1-x^4)} ; \quad x \leq y < 1. \end{aligned}$$

2.6 Independence of Random Variables

Definition : Suppose (X, Y) is a bivariate r.v. with $f(x, y)$ as joint p.d.f., $f_1(x)$ and $f_2(y)$ be the marginal p.d.f. of X and Y respectively. The random variables X and Y are said to be independently distributed iff,

$$f(x, y) = f_1(x) \cdot f_2(y), \quad \forall (x, y).$$

Note : If X and Y are independent then the conditional distribution of X given Y is same as the marginal distribution of X i.e.

$$f_1(x) = g_1(x|y), \quad \forall (x, y)$$

Similarly

$$f_2(y) = g_2(y|x), \quad \forall (x, y).$$

Note :

- (1) If X and Y are independent r.v.s. then,
 $f(x, y) = f_1(x) \cdot f_2(y), \quad \forall (x, y)$
- (2) If X and Y are dependent r.v.s. then,
 $f(x, y) = f_1(x) \cdot g_2(y|x)$ or $f(x, y) = f_2(y) \cdot g_1(x|y)$

Example 2.7 : Verify whether the r.v.s X and Y having p.d.f.

$$f(x, y) = e^{-(x+y)} ; x > 0, y > 0.$$

are independent.

Solution : Marginal p.d.f. of X is given by,

$$f_1(x) = \int_0^{\infty} e^{-(x+y)} dy = e^{-x} \int_0^{\infty} e^{-y} dy = e^{-x} ; x > 0$$

Marginal p.d.f. of Y is given by,

$$f_2(y) = \int_0^{\infty} e^{-(x+y)} dx = e^{-y} \int_0^{\infty} e^{-x} dx = e^{-y} ; y > 0$$

Note that $f(x, y) = e^{-(x+y)} = e^{-x} \cdot e^{-y} = f_1(x) \cdot f_2(y) ; \forall (x, y).$

$\therefore X$ and Y are independent r.v.s.

Example 2.8 : Suppose (X, Y) is a bivariate r.v. with p.d.f.

$$f(x, y) = \begin{cases} 1 & ; \text{ if } 0 < x < 1, -x < y < x \\ 0 & ; \text{ otherwise} \end{cases}$$

Are X and Y independent ?

Solution : The range set is

Here we need to plot the

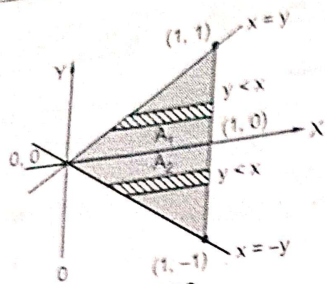


Fig. 2.10

To find the marginal p.d.f. of Y we have to express limits constant and limits of X as functions of Y.

$$A = A_1 \cup A_2 = \{(x, y) \mid 0 < y < 1, y < x < 1\} \cup \{(x, y) \mid -1 < y < 0, -y < x < 1\}$$

$$f_2(y) = \int_{-y}^1 f(x, y) dx \quad ; \quad -1 < y < 0$$

$$= \int_{-y}^1 dx = [x]_{-y}^1 = 1 + y \quad ; \quad -1 < y < 0$$

$$f_2(y) = \int_y^1 dx = [x]_y^1 = 1 - y \quad ; \quad 0 < y < 1$$

$$\therefore f_2(y) = \begin{cases} 1 + y & ; \quad -1 < y < 0 \\ 1 - y & ; \quad 0 < y < 1 \end{cases}$$

Note that $f(x, y) \neq f_1(x) \cdot f_2(y)$

Hence X and Y are dependent r.v.s.

2.7 Expectation, Mean and Moments

Definition : If (X, Y) is a bivariate r.v. with joint p.d.f. $f(X, Y)$, $g(X, Y)$ is a function (X, Y) then expectation of $g(X, Y)$ is denoted $E[g(X, Y)]$ and it is defined as,

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy.$$

provided the double integral exists.

Marginal p.d.f. of X is given by,

$$\begin{aligned} f_1(x) &= \int_{-x}^x f(x, y) dy \\ &= \int_{-x}^x dy = [y]_{-x}^x = x - (-x) \\ &= 2x \quad ; \quad 0 < x < 1 \end{aligned}$$

Note : One can find p.d.f. of $Z = g(X, Y)$ and then find

$$E(Z) = \int_{-\infty}^{\infty} z \cdot h(z) dz \quad ; \quad \text{where } h(z) \text{ is p.d.f. of } z.$$

Moments : To get raw moment of order (r, s) we take

$$g(X, Y) = X^r Y^s$$

$$\mu'_{rs} = E(X^r Y^s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X^r Y^s f(x, y) dx dy.$$

In particular,

$$\text{for } r=1, \quad s=0 \quad ; \quad \text{we get } \mu'_{10} = E(X)$$

$$\text{for } r=0, \quad s=1 \quad ; \quad \text{we get } \mu'_{01} = E(Y)$$

$$\text{for } r=1, \quad s=1 \quad ; \quad \text{we get } \mu'_{11} = E(XY)$$

$$\text{for } r=2, \quad s=0 \quad ; \quad \text{we get } \mu'_{20} = E(X^2)$$

$$\text{for } r=0, \quad s=2 \quad ; \quad \text{we get } \mu'_{02} = E(Y^2)$$

$$\therefore \text{Var}(X) = \mu'_{20} - (\mu'_{10})^2 \quad \text{Var}(Y) = \mu'_{02} - (\mu'_{01})^2$$

$$\text{Cov}(X, Y) = \mu'_{11} - \mu'_{10} \cdot \mu'_{01} \quad E(XY) =$$

Central Moments : If $E(X) = m_1$, $E(Y) = m_2$ then the central moment of order (r, s) is given by,

$$\mu_{rs} = E[(X - m_1)^r (Y - m_2)^s] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (X - m_1)^r (Y - m_2)^s f(x, y) dx dy$$

In particular,

$$\text{for } r=2, \quad s=0 \quad , \quad \mu_{20} = \text{Var}(X)$$

$$\text{for } r=0, \quad s=2 \quad , \quad \mu_{02} = \text{Var}(Y)$$

$$\text{for } r=1, \quad s=1 \quad , \quad \mu_{11} = \text{Cov}(X, Y)$$

$\therefore$ Karl Pearson's coefficient of correlation is,

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = \frac{\mu_{11}}{\sqrt{\mu_{20} \cdot \mu_{02}}}$$

Example 2.9 : Suppose (X, Y) is a bivariate r.v. with p.d.f. (P.U. 2017)

$$f(x, y) = 2 \quad ; \quad 0 < x < y < 1$$

$$= 0 \quad ; \quad \text{otherwise}$$

Find $\text{Corr}(X, Y)$.

Solution : To find means and variances of X and Y we find first marginal p.d.f.s.

The range $A = \{(x, y) \mid 0 < x < y < 1\}$ is equivalent to

$$A_1 = \{(x, y) \mid 0 < x < 1, x < y < 1\} \text{ and}$$

$$A_2 = \{(x, y) \mid 0 < y < 1, 0 < x < y\}$$

To get the marginal p.d.f. of X we use A_1

$$f_1(x) = \int_x^1 f(x, y) dy = \int_x^1 2 dy = 2(1-x); \quad 0 < x < 1$$

and the marginal p.d.f. of Y is given by,

$$f_2(y) = \int_0^y x \cdot f(x, y) dx = \int_0^y 2x dx = 2y; \quad 0 < y < 1$$

Clearly,

$$E(X) = \int_0^1 x f_1(x) dx = \int_0^1 x \cdot 2(1-x) dx = 1/3$$

$$E(X^2) = \int_0^1 x^2 \cdot 2(1-x) dx = 1/6$$

$$E(Y) = \int_0^1 y f_2(y) dy = \int_0^1 2y^2 dy = 2/3$$

$$E(Y^2) = \int_0^1 y^2 f_2(y) dy = \int_0^1 2y^3 dy = 1/2$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = 1/18$$

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = 1/18$$

To find $\text{cov}(X, Y)$ first we find $E(XY)$

$$E(XY) = \int_0^1 \int_0^y xy \cdot f(x, y) dx dy = 2 \int_0^1 \int_0^y xy dx dy$$

$$= 2 \int_0^1 y [x^2/2]_0^y dy = \int_0^1 y^3 dy = \frac{1}{4}$$

$$\text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y) = \frac{1}{4} - \frac{1}{3} \cdot \frac{2}{3} = \frac{1}{36}$$

$$\rho = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = \frac{1/36}{\sqrt{\frac{1}{18} \cdot \frac{1}{18}}} = \frac{1}{2}$$

Remark : Without obtaining marginal p.d.f. one can directly find the means and variances as follows :

$$E(X) = \int_0^1 \int_0^y x \cdot f(x, y) dx dy$$

$$= \int_0^1 \left[\int_0^y 2x dx \right] dy = \int_0^1 y^2 dy$$

$$= \left[\frac{y^3}{3} \right]_0^1 = 1/3.$$

Similarly, $E(X^2) = \int_0^1 \int_0^y x^2 f(x, y) dx dy = \int_0^1 \frac{2}{3} y^3 dy = \frac{1}{6}$

and $E(Y) = \int_0^1 \int_0^y y f(x, y) dx dy = \int_0^1 2y^2 dy = \frac{2}{3}$ and

and $E(Y^2) = \int_0^1 \int_0^y y^2 f(x, y) dx dy = \int_0^1 2y^3 dy = \frac{1}{2}.$

Then as discussed above we can find $\text{corr}(X, Y)$.

Theorems on Expectation :

Theorem 1 : If (X, Y) is a bivariate r.v. with joint p.d.f. $f(x, y)$ then,

$$E(X + Y) = E(X) + E(Y)$$

Proof: By definition,

$$\begin{aligned}
 E(X+Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f(x,y) dx dy \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x,y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x,y) dx dy \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x,y) dy dx + \int_{-\infty}^{\infty} y \left[\int_{-\infty}^{\infty} f(x,y) dx \right] dy \\
 &= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f(x,y) dy \right] dx + \int_{-\infty}^{\infty} y \cdot f_2(y) dy \\
 &= \int_{-\infty}^{\infty} x \cdot f_1(x) dx + \int_{-\infty}^{\infty} y f_2(y) dy = E(X) + E(Y)
 \end{aligned}$$

Here $f_1(x)$ is a marginal p.d.f. of X and $f_2(y)$ is a marginal p.d.f. of Y .

Note: We can prove by similar way a more general result

$$E(aX + bY + c) = aE(X) + bE(Y) + c$$

where a, b, c are constants.

If $a = b = 1$ then, we get $E(X+Y) = E(X) + E(Y)$ and for $a = 1, b = -1$, we get $E(X-Y) = E(X) - E(Y)$.

Theorem 2: If (X, Y) is a bivariate r.v. with joint p.d.f. $f(x, y)$ and X, Y are independently distributed then,

$$E(XY) = E(X)E(Y)$$

Proof: By definition

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x,y) dx dy$$

Since, X and Y are independent r.v.s. we get,

$$f(x, y) = f_1(x) \cdot f_2(y), \forall (x, y)$$

where $f_1(x)$ and $f_2(y)$ are marginal p.d.f.s.

$$\begin{aligned}
 E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_1(x) f_2(y) dx dy \\
 &= \int_{-\infty}^{\infty} y f_2(y) \left[\int_{-\infty}^{\infty} x f_1(x) dx \right] dy \\
 &= \int_{-\infty}^{\infty} y f_2(y) E(X) dy = E(X) \int_{-\infty}^{\infty} y f_2(y) dy \\
 &= E(X) \cdot E(Y).
 \end{aligned}$$

As usual we summarise some more results.

Result: If X and Y are independent r.v.s. then they are uncorrelated.

Proof: Since, X and Y are independent,

$$E(XY) = E(X) \cdot E(Y)$$

$$E(XY) - E(X) \cdot E(Y) = 0$$

$$\text{Cov}(X, Y) = 0$$

$\therefore X$ and Y are uncorrelated.

Remark: Converse of the above result does not hold.

Counter example: Suppose, X and Y are identically distributed random variables. Define $U = X + Y$ and $V = X - Y$.

$$\begin{aligned}
 \text{Clearly, } E(U \cdot V) &= E[(X+Y)(X-Y)] = E(X^2 - Y^2) \\
 &= E(X^2) - E(Y^2)
 \end{aligned}$$

($\because X$ and Y are identically distributed)

$$= 0$$

and

$$E(V) = E(X - Y) = E(X) - E(Y) = 0$$

$$\therefore \text{Cov}(U, V) = E(UV) - E(U) \cdot E(V) = 0$$

$\therefore U$ and V are uncorrelated. However, $U = X + Y$ and $V = X - Y$ need not be independent.

2.8 Moment Generating Function (M.G.F.)

Moment generating function of (X, Y) helps in finding moments of X , moments of Y as well as joint moments of X and Y . Moreover, we can verify whether X and Y are independent using M.G.F. Sometimes M.G.F. can be used to find probability distribution of function of X and Y .

Definition : If (X, Y) is a bivariate r.v. then the M.G.F. of (X, Y) denoted by $M_{X,Y}(t_1, t_2)$ and it is defined as,

$$M_{X,Y}(t_1, t_2) = E[e^{t_1 X + t_2 Y}]$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x + t_2 y} f(x, y) dx dy$$

In order to find the moments we expand M.G.F. as follows:

$$M_{X,Y}(t_1, t_2) = 1 + (t_1 \mu'_{10} + t_2 \mu'_{01})$$

$$+ \frac{1}{2!} (t_1^2 \mu'_{20} + 2 t_1 t_2 \mu'_{11} + t_2^2 \mu'_{02})$$

$$+ \frac{1}{3!} (t_1^3 \mu'_{30} + 3 t_1^2 t_2 \mu'_{21} + 3 t_1 t_2^2 \mu'_{12} + t_2^3 \mu'_{03})$$

$$+ \dots + \dots$$

μ'_{rs} = coefficient of $\frac{t_1^r t_2^s}{r! s!}$ in the expansion of $M_{X,Y}$

μ'_{rs} can also be obtained as $\left[\frac{\partial^{r+s}}{\partial t_1^r \partial t_2^s} M_{X,Y}(t_1, t_2) \right]_{(0,0)}$

Remarks : (1) $M_{X,Y}(0, 0) = 1$

(2) Using joint M.G.F. of (X, Y) we can get M.G.F. of X as follows

$$M_{X,Y}(t_1, 0) = E(e^{t_1 X}) = M_X(t_1).$$

Similarly,

$$M_{X,Y}(0, t_2) = E(e^{t_2 Y}) = M_Y(t_2).$$

Theorem 3 : Suppose (X, Y) is a continuous bivariate r.v. with p.d.f. $f(x, y)$.

X and Y are independent if and only if

$$M_{X,Y}(t_1, t_2) = M_{X,Y}(t_1, 0) \cdot M_{X,Y}(0, t_2).$$

Part (1) : Here, under the assumption that X and Y are independent we prove $M_{X,Y}(t_1, t_2) = M_{X,Y}(t_1, 0) \cdot M_{X,Y}(0, t_2)$. Note that,

$$M_{X,Y}(t_1, t_2) = E[e^{t_1 X + t_2 Y}]$$

$$= E[e^{t_1 X} \cdot e^{t_2 Y}] \quad (\because X \text{ and } Y \text{ are independent})$$

$$= E(e^{t_1 X}) \cdot E(e^{t_2 Y})$$

$$= M_{X,Y}(t_1, 0) \cdot M_{X,Y}(0, t_2).$$

Part (2) : Here we assume

$M_{X,Y}(t_1, t_2) = M_{X,Y}(t_1, 0) \cdot M_{X,Y}(0, t_2)$ and we want to prove that X and Y are independent.

Note that,

$$M_{X,Y}(t_1, t_2) = M_{X,Y}(t_1, 0) \cdot M_{X,Y}(0, t_2)$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x + t_2 y} f(x, y) dx dy = \int_{-\infty}^{\infty} e^{t_1 x} f_1(x) dx \int_{-\infty}^{\infty} e^{t_2 y} f_2(y) dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x + t_2 y} f_1(x) f_2(y) dx dy$$

$$f(x, y) = f_1(x) \cdot f_2(y); \forall (x, y).$$

$\therefore$ Thus X and Y are independent random variables.

Example 2.10 : If (X, Y) is a r.v. with p.d.f.

$$f(x, y) = e^{-(x+y)} \quad ; \quad x > 0, y > 0$$

$$= 0 \quad ; \quad \text{otherwise.}$$

Find M.G.F. of (X, Y) . Hence, find M.G.F. of X , M.G.F. of Y . Verify whether X and Y are independent. Obtain the M.G.F. of $X + Y$ using joint M.G.F.

Solution : (i) $M_{X,Y}(t_1, t_2) = \int_0^{\infty} \int_0^{\infty} e^{t_1 x + t_2 y} e^{-(x+y)} dx dy$

$$= \int_0^{\infty} e^{-(1-t_1)x} dx \cdot \int_0^{\infty} e^{-(1-t_2)y} dy$$

$$= \frac{1}{1-t_1} \times \frac{1}{1-t_2} = \frac{1}{(1-t_1)(1-t_2)}$$

(ii) M.G.F. of $X = M_{X,Y}(t_1, 0) = \frac{1}{(1-t_1)(1-0)} = \frac{1}{1-t_1}$

(iii) M.G.F. of $Y = M_{X,Y}(0, t_2) = \frac{1}{(1-0)(1-t_2)} = \frac{1}{1-t_2}$

(iv) $M_{X,Y}(t_1, t_2) = M_{X,Y}(t_1, 0) \cdot M_{X,Y}(0, t_2)$.

$\therefore$ X and Y are independent r.v.s.

(v)

$$\text{M.G.F. of } (X + Y) = M_{X,Y}(t, t)$$

$$M_{X+Y}(t) = \frac{1}{(1-t)} \cdot \frac{1}{(1-t)} = \frac{1}{(1-t)^2}$$

put $t_1 = t$

2.9 Conditional Expectation

In the study of bivariate probability distributions, we are interested in (i) the marginal distribution of each r.v. (ii) the distribution of one variable when the other is fixed at a certain point in its range, which is called as conditional distributions. Marginal distribution is free from one variable. It is like a usual univariate r.v. Whereas in conditional distribution, the value of other variable is involved in p.d.f. so also in the moments. However, in case the r.v.s. are independent then conditional distribution is same as marginal distribution of corresponding r.v. Here we find mean and variance of a variable using conditional probability distribution.

Definition : Let (X, Y) be a bivariate r.v. with joint p.d.f. $f(x, y)$. Suppose, $g_1(x|y)$ is a conditional p.d.f. of X given y . The conditional expectation of X given y is denoted by $E(X|y)$ and it is defined as,

$$E(X|y) = \int_{-\infty}^{\infty} x \cdot g_1(x|y) dx$$

Similarly,

$$E(Y|x) = \int_{-\infty}^{\infty} y \cdot g_2(y|x) dy$$

Where, $g_2(y|x)$ is a conditional p.d.f. of Y given x . The second order moments and variance using conditional distributions will be,

$$E(X^2|y) = \int_{-\infty}^{\infty} x^2 \cdot g_1(x|y) dx \text{ and}$$

$$\text{Var}(X|y) = E(X^2|y) - [E(X|y)]^2$$

Similarly,

$$E(Y^2|x) = \int_{-\infty}^{\infty} y^2 \cdot g_2(y|x) dy, \text{ and}$$

$$\text{Var}(Y|x) = E(Y^2|x) - [E(Y|x)]^2$$

Remark (1) : $E(X|y)$ may not be a constant like $E(X)$, it will be a function of r.v. Y . Thus, $E(X|y) = \psi(y)$ will itself be a r.v. Similarly $E(Y|x)$ is a function of r.v. x . We can obtain its expectation as follows :

It is interesting to note that

Remark (2) : $E(E(X|y)) = E(X)$ and $E(E(Y|x)) = E(Y)$.

Proof :

$$E(E(X|y)) = E(h(y)) \quad \text{since } E(X|y) = h(y) \text{ is a function of } y$$

$$= \int_{-\infty}^{\infty} h(y) \cdot f_2(y) dy = \int_{-\infty}^{\infty} E(X|y) f_2(y) dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \cdot g_1(x|y) f_2(y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dy dx$$

$$= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f(x, y) dy \right) dx = \int_{-\infty}^{\infty} x f_1(x) dx = E(X)$$

Remark (3) : Correlation coefficient and conditional expectation :

Note that $E(X|y) = \psi(y)$ is a some function of y . If it is a linear function of y then the coefficient of Y is a regression coefficient of X on Y .

Suppose $E(X|y) = a + by$, then, the regression coefficient of X on Y is b .

Similarly, if $E(Y|x) = c + dx$, then, the regression coefficient of Y on X is d .

$$\text{Therefore, } \text{corr}(X, Y) = \sqrt{bd}$$

Remark (4) : It may happen $E(X|y)$ is a linear function of y and $E(Y|x)$ is a non-linear function of x or vice-versa. In this case we cannot find correlation coefficient between X and Y . Moreover, it is irrelevant to study the same because regression is non-linear.

Example : Suppose (X, Y) is a bivariate r.v. with joint p.d.f.

$$f(x, y) = 1 \quad ; \quad 0 < x < 1, \quad -x < y < x \\ = 0 \quad ; \quad \text{otherwise.}$$

Solved Examples

Example 2.11 : Let (X, Y) have joint p.d.f.

$$f(x, y) = 2 \quad ; \quad 0 < x < y < 1 \\ = 0 \quad ; \quad \text{otherwise.}$$

Find (i) conditional distribution of X given y , (ii) conditional distribution of Y given x , (iii) $E(X|y)$, $E(Y|x)$ hence or otherwise find $\text{Cov}(X, Y)$. (iv) Verify $E(E(X|y)) = E(X)$.

Solution : To find conditional distributions we need to find first of all marginal distributions. Let, us express the range set A suitably to get marginal distributions.

$A = \{(x, y) | 0 < x < y < 1\}$, $A_1 = \{(x, y) | 0 < x < 1, x < y < 1\}$
 $A_2 = \{(x, y) | 0 < y < 1, 0 < x < y\}$. Clearly $A = A_1 = A_2$.

The marginal p.d.f. of X using A_1 is,

$$f_1(x) = \int_x^1 f(x, y) dy = 2(1-x),$$

and the marginal p.d.f. of Y using A_2 is,

$$f_2(y) = \int_0^y f(x, y) dx = 2y,$$

(i) conditional distribution of X given y is

$$g_1(x|y) = \frac{f(x, y)}{f_2(y)} = \frac{2}{2y} = \frac{1}{y},$$

$$E(X|y) = \int_0^y x g_1(x|y) dx = \frac{1}{y} \int_0^y x dx = \frac{y}{2}.$$

(ii) Conditional distribution of Y given x is

$$g_2(y|x) = \frac{f(x, y)}{f_1(x)} = \frac{2}{2(1-x)} = \frac{1}{1-x}.$$

$$E(Y|x) = \int_x^1 y g_2(y|x) dy = \int_x^1 \frac{y}{1-x} dy = \frac{1}{1-x} \left[\frac{y^2}{2} \right]_x^1$$

$$= \frac{1}{1-x} \cdot \frac{1-x^2}{2} = \frac{1+x}{2}.$$

(iii) Since, $E(X|y) = \frac{y}{2}$ is a linear,

$$b_{xy} = \text{Regression coefficient of X on Y} \\ = \text{Coefficient of y in } E(X|y) = 1/2.$$

Similarly, due to $E(Y|x) = (1+x)/2$ we get,

$$b_{yx} = \text{Regression coefficient of Y on X} \\ = \text{Coefficient of x in } E(Y|x) = 1/2.$$

$$\therefore \text{Corr}(x, y) = \sqrt{b_{xy} \cdot b_{yx}} = \sqrt{\frac{1}{2} \cdot \frac{1}{2}} = \frac{1}{2}.$$

$$(iv) E(X|y) = \frac{y}{2} \therefore E(E(X|y)) = E\left(\frac{y}{2}\right) = \frac{1}{2} \int_0^1 y f_2(y) dy$$

$$= \frac{1}{2} \int_0^1 2y^2 dy = \left[\frac{y^3}{3} \right]_0^1 = \frac{1}{3}$$

$$E(X) = \int_0^1 x f_1(x) dx = 2 \int_0^1 x(1-x) dx$$

$$= 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3} \therefore E(E(X|y)) = E(X)$$

Example 2.12 : Suppose (X, Y) is a bivariate r.v. The marginal distribution of X is given by $f_1(x) = \frac{2x+1}{2}$, $0 < x < 1$ and the conditional distribution of Y given x is

$$g_2(y|x) = \frac{2(x+y)}{2x+1}; \quad 0 < y < 1$$

Find, (i) marginal distribution of Y.

(ii) conditional distribution of X given y.

(iii) Are X and Y independently distributed?

Solution : The joint p.d.f. of X will be,

$$f(x, y) = f_1(x) \cdot g_2(y|x) = \frac{2x+1}{2} \cdot \frac{2(x+y)}{2x+1}$$

$$= (x+y) \quad \text{for } 0 < x < 1, 0 < y < 1.$$

(i) To get marginal distribution of Y, we evaluate,

$$f_2(y) = \int_0^1 f(x, y) dx = \int_0^1 (x+y) dx = \left[\frac{x^2}{2} + xy \right]_0^1$$

$$= y + \frac{1}{2}, \quad 0 < y < 1.$$

(ii) Conditional distribution of X given y will be,

$$g_1(x|y) = \frac{f(x, y)}{f_2(y)} = \frac{x+y}{y + \frac{1}{2}} = \frac{2(x+y)}{2y+1}, \quad 0 < y < 1.$$

(iii) Since, $f(x, y) = x+y$ and $f_1(x) \cdot f_2(y) = \left(x + \frac{1}{2}\right) \left(y + \frac{1}{2}\right)$

Therefore, $f(x, y) \neq f_1(x) \cdot f_2(y)$

Hence, X and Y are dependent r.v.s.

2.10 Transformations of Bivariate Continuous Random Variables

Often we are interested in finding probability distribution functions such as, $X + Y$, $X - Y$, XY , X/Y , $X/(X + Y)$, $Y/(X + Y)$, $X^2 + Y^2$. Where (X, Y) is a bivariate r.v. The various methods which are practised for this purpose are as follows :

- (i) Using distribution function.
- (ii) Finding p.d.f. using transformation method.
- (iii) Using M.G.F.

The choice of method depends upon the nature of transformation as well as distribution of (X, Y) .

Method (i) : Whenever it is possible we can find distribution function and differentiating it we get p.d.f.

This method is not suitable for every transformation.

Note : (1) Some times we are interested in only one transformation such as $U = X + Y$, $U = X - Y$, $U = X/(X + Y)$ then the variable V is chosen as $V = Y$ or $V = X$. The choice $V = Y$ or $V = X$ depends upon convenience in evaluation of marginal p.d.f. of U i.e. $h_1(u)$.

Example 2.14 : Suppose X , Y , are independent random variables with p.d.f.s as $f_1(x) = e^{-x}$, $x > 0$ and $f_2(y) = e^{-y}$, $y > 0$. Find the probability distributions of $X + Y$ and $\frac{X}{Y}$.

Solution : **Step (1) :** Define the transformations. Let, $U = X + Y$, $V = \frac{X}{Y}$. The range will be $0 < U < \infty$ and $0 < V < \infty$.

Step (2) : Express X and Y in terms of U and V . (Inversion transformation)

Note that, $V = \frac{X}{Y}$,
 $U = X + Y = VY + Y = (1 + V)Y$
 $\therefore Y = \frac{U}{1 + V}$
 $X = \frac{UV}{1 + V}$

Since, $V = \frac{X}{Y}$, we get $X = VY = \frac{UV}{1 + V}$

$\therefore X = \frac{UV}{1 + V}$ and $Y = \frac{U}{1 + V}$

Step (3) : Find $J = \frac{\partial(X, Y)}{\partial(U, V)}$

$$J = \begin{vmatrix} \frac{\partial X}{\partial U} & \frac{\partial X}{\partial V} \\ \frac{\partial Y}{\partial U} & \frac{\partial Y}{\partial V} \end{vmatrix} = \begin{vmatrix} \frac{V}{1 + V} & \frac{U}{(1 + V)^2} \\ \frac{1}{1 + V} & \frac{-U}{(1 + V)^2} \end{vmatrix}$$

$$= \frac{1}{(1 + V)^3} \cdot (-UV - U) = -\frac{U}{(1 + V)^2}$$

Step (4) : Obtain joint p.d.f. of (U, V)

$h(u, v) = f(x, y) \cdot |J| = f_1(x) \cdot f_2(y) \cdot |J|$ ($\because X$ and Y are independent)

$\int \int f(x, y) dx dy$

$$h(u, v) = e^{-(x+y)} \cdot |J| ; x > 0, y > 0$$

$$= e^{-u} \cdot \frac{u}{(1 + v)^2} ; u > 0, v > 0$$

Step (5) : Find marginal p.d.f.s

p.d.f. of U will be $h_1(u) = \int_0^\infty h(u, v) dv$

$$h_1(u) = \int_0^\infty e^{-u} u \frac{1}{(1 + v)^2} dv, u > 0$$

$$= u e^{-u} [-1(1 + v)^{-1}]_0^\infty = u e^{-u}, u > 0$$

Similarly, p.d.f. of V will be,

$$h_2(v) = \int_0^\infty h(u, v) du = \int_0^\infty e^{-u} u \frac{1}{(1 + v)^2} du$$

$$h_2(v) = \frac{1}{(1 + v)^2} \int_0^\infty e^{-u} u du$$

$$h_2(v) = \frac{2}{(1 + v)^2} ; u > 0$$

Alternatively, in this problem we can express $h(u, v)$ as product of function of u and function of v . These functions turn out to be p.d.f. of r.v.s U and V .

$$h(u, v) = (e^{-u} \cdot u) \cdot \left(\frac{1}{(1 + v)^2} \right)$$

$$= h_1(u) \cdot h_2(v)$$

Therefore, U and V are independent r.v.s.

Note : In order to find the distribution of $X + Y$ only, we can define $U = X + Y$ and $V = Y$ and use the above method.

Similarly to get the distribution of $\frac{X}{Y}$, we can define $U = \frac{X}{Y}$ and $V = Y$ and use the above method.